TSDL List of Experiments

Sr. No.	Assignment
1	Installation of Java and Eclipse IDE
2	Design a class named "Product" with default, and parameterized constructors. a. Default constructor initializes default values b. the parameterized constructor allows custom initialization Implement methods to display product details. Test the class by creating instances using each constructor to make sure everything works correctly and data is handled properly. creating multiple product objects, store their data in a suitable collection (such as an ArrayList) and perform basic data analysis on the product dataset.
3	Create a base class Person with attributes name and age. Include a method display_info() to print basic information about the person. **a. Single Inheritance:** Derive a class Student from the Person class with additional attributes student_id and major. Include a method enroll_course() to simulate a student enrolling in a course. **b. Multilevel Inheritance:** Extend the Student class to create a new class GraduateStudent with an additional attribute research_topic. Implement a method conduct_research() to simulate a graduate student conducting research. **c. Hierarchical Inheritance:** Derive a class Professor from the Person class with attributes like employee_id, department, and teaching_subject. Implement a method conduct_lecture() to simulate a professor conducting a lecture. Design a multilevel inheritance system to handle dataset processing. Implement a Data Cleaning Algorithm to remove invalid records such as negative marks or empty names.
4	Multiple inheritance using Interface: **a) Define two interfaces:** CardioWorkout and StrengthTraining. In the CardioWorkout interface, declare methods such as getWorkoutName(), getDurationMinutes(), and displayWorkoutDetails(). b) In the StrengthTraining interface, declare methods like getEquipmentUsed(), getInstructorName(), and displayStrengthTrainingDetails(). **c) Derived Class:** Create a class named FullBodyWorkout that implements both Cardio_Workout and StrengthTraining. This class represents a workout program that combines cardio exercises with strength training for a comprehensive full-body workout. Design a Java program using interfaces to implement a simple AI-based recommendation system. Create interfaces such as UserPreference and RecommendationEngine. The program should store product or course data and recommend suitable items to users based on predefined rules such as price range, category, or rating.
Lab CIE 1	

5	Create a base class Account featuring methods deposit() and withdraw(). These methods should be overridden by derived classes to execute specific actions tailored to different account types. **a. Method Overloading:** Derive a class SavingsAccount from Account. Implement method overloading for withdraw() i.Illustrate polymorphism by invoking the withdraw() method on both objects, highlighting distinct behaviors based on the account type. by introducing an additional parameter to specify the reason for withdrawal. Create a Java program using **method overloading** to implement a **Data Categorization Algorithm**. **b. Method Overriding:** Derive a class CheckingAccount from Account. Implement method overriding for withdraw() within the CheckingAccount class, introducing overdraft protection. **c. Polymorphic Demonstration:** i) Instantiate objects for both SavingsAccount and CheckingAccount. ii) Illustrate polymorphism by invoking the withdraw() method on both objects, highlighting distinct behaviors based on the account type. Develop a Java program using inheritance to demonstrate **method overriding**. Create a base class Account and a derived class SavingsAccount that overrides the withdraw() method. Implement a **Rule-Based Classification Algorithm** to categorize transactions as "Approved" or "Denied" based on balance and minimum balance rules.
6	Write a Java program in which you will declare an abstract class Vehicle inherits this class from two classes car and truck using the method engine in both displays "car has good engine" and "truck has bad engine". Write a Java program to demonstrate abstraction and inheritance by creating a general vehicle structure and extending it into Car and Truck with different engine behavior. Assign a performance score to each vehicle. Apply a **Rule-Based Classification Algorithm** to categorize vehicles as High, Medium, or Low Performance. Display the vehicle details along with performance score and category.
7	Develop an Employee Management System and demonstrate the concept of encapsulation. Create a class named Employee with private attributes for employeeId, firstName, lastName, dateOfBirth, and salary. Implement accessor methods for retrieving attribute values and mutator methods for modifying attributes, ensuring proper validation. Additionally, include a method calculateYearlyBonus() within the Employee class that utilizes encapsulated data to calculate a yearly bonus based on a predetermined formula.
8	Create a Java class named "Reservation" to manage hotel room reservations. Develop methods to the reservation details from the user and print it. a) Implement a try-and-catch block to handle potential errors that may arise during reservation initialization and display, such as negative guest numbers or null reservation dates. Print relevant error messages. b) Use multiple catch statements to handle distinct types of exceptions, like negative room numbers or null guest names. Handle these exceptions separately during both initialization and display operations. c) Introduce a finally block to ensure cleanup operations, guaranteeing proper resource management regardless of whether an exception occurred. Verify the class by testing instances created using each constructor to confirm correct exception handling and resource cleanup in a hotel room reservation system.
Lab CIE 2	

9	Collections - Arraylist Design a Java program to manage course enrollment details using an ArrayList. Implement the following functionalities using built-in methods(such as add(), set(), size(), remove(), clear(),contains(),isEmpty()) and user input: a. Include course_id, course name, students enrolled, student_name, and student_id. b.Use built-in methods to add this entry to the ArrayList. b. Display a success message after adding the entry. Add a feature to count how many students are enrolled in each course using a **Frequency Count Algorithm**.
10	Topic: Flight Management system (for a particular airline of your choice) Create a system for monitoring flight status tracking, encompassing the following functionalities: a) Departure flights a) Arrival Flight b) Delayed flights c)Cancelled flights
13	Mini Project